

NexForge Automation — Delivery & Workflow Guide

VPD Technologies

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NexForge Automation — Project Delivery & Workflow Guide

A plain-language reference for explaining the NexForge website: how it works end to end, what was built **from the client content pack**, and what was **added by us** on top. Use this to brief the team and management.

1. Project at a glance

Item	Detail
What it is	Enterprise marketing website + enquiry-to-lead CRM
Stack	Python / Django 6, Django REST Framework, PostgreSQL (Supabase), Bootstrap 5, vanilla JS
Surfaces	Public website (/), REST API (/api/v1/), Admin/CRM (/admin/)
Pages	14 public pages, all responsive
Content	14 services, 9 projects, 7 blog posts, 13 FAQs, 4 awards, 7 downloads, testimonials
Quality	55 automated tests passing, 0 browser console errors, security-hardened

2. How the whole site works (paragraph summary)

The site is **one Django application talking to one PostgreSQL database**, exposed through three doors. The **public website** is for anonymous visitors — they read content (services, projects, blog, etc.) and can submit the **contact form**. They never log in. When a visitor submits the form, the data is sent to the **REST API**, which saves it as an **Enquiry (a sales lead)** in the database and emails the sales inbox. Staff use the **Admin panel** (the back door) to manage every piece of content and to work the leads (change their status: New → In progress → Converted → Closed). That is the complete

business loop: *visitor browses* → *submits enquiry* → *lead is created* → *sales follows up*.
There is no visitor login or signup; individual visitors are anonymous, and only the people who fill the form become trackable leads.

2.1 REQUEST LIFECYCLE (WHAT HAPPENS ON EVERY CLICK)

```
Browser → (prod: Cloudflare → Nginx) → Gunicorn → Django middleware
      → URL router → View → Models (ORM) → PostgreSQL
      → HTML page OR JSON → back to Browser
```

2.2 THE LEAD WORKFLOW (THE ONLY FLOW THAT WRITES DATA)

```
Visitor fills Contact form
  | (JavaScript fetch)
  ▼
POST /api/v1/enquiries/      ← rate-limited 5/min, validated
  |
  ├── Save Enquiry row (status = New) ← SOURCE OF TRUTH
  └── Email the sales inbox (fail-safe: lead saved even if email fails)
  ▼
Admin ► Enquiries ► assign + change status ► follow up manually
```

2.3 WHO CAN DO WHAT (ROLES)

```
Anonymous visitor → browse, submit enquiry
Staff user (login) → manage content + leads, scoped by their Role group:
  Administrator → everything
  Content Manager → projects, services, blog, content
  Sales Manager → enquiries (leads)
  Technical Support → enquiries
  HR Manager → (reserved for Careers – not built yet)
```

Roles are Django **Groups**. They exist in Admin → *Groups*. They only take effect once you create staff users and assign each to a group. The current single login is a **superuser**, which sees everything (that is why only “administrator” appears so far).

2.4 CONTENT FLOW (HOW THE SITE STAYS UP TO DATE)

Editor logs into /admin/ → adds/edits a Project (with images, timeline, etc.)
→ saved to database → instantly live on the website + API

No code change or redeploy needed to update content.

3. Implemented FROM the client content pack (the brief)

Everything the master content pack specified that is **built and working**:

Area	Delivered
Company info, vision, mission, values	About page + footer
Brand (colours #0B2E59 / #F57C00 / #00BCD4, Poppins + Inter)	Applied site-wide
Homepage sections	Sticky nav, hero, about, business statistics, services, industries, process, featured projects
Hero banner	Exact heading + sub-heading + 4 CTAs
Business statistics	All 10 stats (animated counters)
Services	All 10 (14 delivered), list + detail pages
Projects	Portfolio with filters (status / industry / technology), search, grid-list toggle, lightbox; ongoing + completed; detail pages with value, status, duration, client, overview, challenges, solution, technologies, deliverables, timeline, related projects
Blog	Featured + recent posts, 8 categories, list + detail
FAQ	General + service questions, accordion
Contact	Form (saves lead + emails sales), contact info; map slot wired (needs maps key)
Gallery / Awards / Downloads	All three pages
Privacy Policy / Terms & Conditions	Full content pages
Admin panel	Django Admin: content management, user management, role-based access
User roles	All 5 roles as Groups

Area	Delivered
Database tables	Users, Projects, Services, Clients, Blog, Testimonials, Enquiries (+ more)
APIs	DRF endpoints: projects, services, blog, enquiries, content, auth

4. Added BY US (beyond the brief — value engineering)

Things not explicitly in the content pack that we added for quality, security, and a premium enterprise feel:

Security & reliability

- Content Security Policy (nonce-based) + HSTS, secure cookies, X-Frame-Options, no-sniff
- API rate-limiting (anti-spam on the contact form; brute-force guard on login)
- Server-side file-upload validation (type + size)
- Per-user data access on leads (prevents one staff user seeing another's, IDOR-safe)
- `.env`-based secrets, split dev/prod settings, dependency CVE scan (clean)
- Custom error pages (400 / 403 / 404 / 500)

Developer & operations

- Auto-generated API documentation (Swagger UI at `/api/docs/`)
- SEO: `sitemap.xml` + `robots.txt`
- Seed command (one command loads all demo content)
- `run.bat` one-click launcher, README, automated test suite (55 tests)
- Google Analytics hook (turns on by setting one env value)

Premium UI / motion layer

- Hero entrance animation, floating hero graphic, animated stat counters
- Scroll-reveal across the whole site (cards, sections, lists)

- Button shine-sweep, card hover glow, animated section underlines
- Technology-partner marquee (auto-scrolling logo strip)
- Custom admin dashboard with live count tiles + recent enquiries
- Branded gradient placeholders for items without photos
- Full accessibility: respects “reduced motion”, keyboard focus rings

5. Not yet built (Phase 2 — honest scope)

These appear in the broader business-workflow vision but are **not in the current build**:

Module	Status	Note
Careers / Recruitment (jobs, resume upload, HR review)	Not built	HR role exists; needs a <code>careers</code> app. Recommended placement: footer + <code>/careers/</code>
Support / Ticketing	Not built	No ticket system in the brief or build
Customer login / portal	Not built	Decision pending with management; visitors are anonymous today
Async notifications (Celery + Redis + signals)	Not built	Email is currently sent synchronously, sales-only
Customer auto-reply email	Not built	Submitter sees on-screen thanks only
JWT authentication	Not built	Uses DRF Token auth instead
Sales pipeline (stages, deal value, activity log)	Partial	Only 4 lead statuses today
Analytics charts / reports	Partial	Count tiles only; no graphs
Real photography	Pending	Gradient placeholders until real images supplied

6. Operating notes

- **Run locally:** double-click `run.bat`, open `http://127.0.0.1:8000/`.
- **Admin login:** create one with `python manage.py createsuperuser`.
- **Before production:** rotate the database password, set `DEBUG=False` + `config.settings.prod`, configure real SMTP, add Google Maps + Analytics keys, run `collectstatic`, supply real images.

Document generated as a delivery reference. Keep in sync with the codebase.